Nasim Paykari

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Research Interests

Cooperative multi-robot navigation; map-free visual navigation in GPS-denied environments; computer vision; cross-view landmark recognition; agentic AI for wide-area visual homing; distributed systems, blockchain consensus; cyber-physical systems; industrial automation and control.

Education

Doctor of Philosophy (Ph.D.) in Computer Science – Candidate
Fordham University
2022–May 2027 (expected)

Dissertation: *Integrating Blockchain and Wide Area Visual Navigation for Secure, Decentralized Autonomous Robotic Systems*.
Co-advisers: Damian M. Lyons and Mohamed Rahouti.
Cumulative GPA: 3.958/4.000.

Master of Science (M.Sc.) in Instrumentation and Automation Engineering
Shiraz University
2013

Specialization: Instrumentation Engineering and Automation in Oil Industries.
Thesis: *Design of Fuzzy Controller for Wall Following Mobile Robot with Stability Analysis*. Adviser: F. Shabaninia.

Bachelor of Science (B.Sc.) in Electrical Engineering – Electronics
Lorestan University
September 2009

Research Experience

Doctoral Researcher
Robotics and Computer Vision Laboratory, Fordham University
During Ph.D. studies, 2022–present

Co-advisers: Damian M. Lyons and Mohamed Rahouti.

- Currently investigate agentic artificial intelligence for wide-area visual homing.

- Conduct research on Wide Area Visual Navigation (WAVN), enabling cooperative robot teams to navigate toward out-of-view destinations using shared visual landmarks in GPS-denied environments.
- Develop and evaluate blockchain-based coordination and consensus methods for cooperative visual homing, including RoboStake and Witness Byzantine Fault Tolerance with Proof-of-Navigation.
- Investigate decentralized pathfinding and route recovery on shared landmark graphs, including alternative-path strategies for changing connectivity and navigation failures.
- Study cross-view landmark recognition across widely separated viewpoints to identify common landmarks observed by different robots.
- Implement and evaluate navigation experiments primarily in ROS and Gazebo, with additional work on Turtle robot platforms.

Master's Thesis Researcher
Shiraz University
2010–2013

- Investigated fuzzy control for wall-following mobile robots, including MIMO Mamdani controllers and stability analysis.
- Coauthored a research paper published in the proceedings of the Soft Computing Applications workshop.

Research Proposal Development

Research Proposal Contributor – National Science Foundation (NSF)
Fordham University
During doctoral research

Contributed to the technical development and writing of a faculty-led NSF proposal on cooperative robotic navigation, drawing on doctoral research in map-free visual navigation and decentralized coordination. Collaborated with faculty investigators to articulate the proposed research approach and its technical rationale.

Teaching Experience

Instructor of Record – Programming with Python (CISC 5380)
Fordham University
Spring 2026

- Sole instructor for an interdisciplinary class of six students, including bachelor's and master's students in computer science and Ph.D. students from other departments, including psychology and finance.
- Responsible for the syllabus, lectures, assignments, examinations, projects, and grading.
- Taught Python programming, functions, object-oriented programming, modular design, and exception handling, with practical programming assignments and an applied final project.

Instructor of Record – Structures of Computer Science (CISC 1100)
Fordham University
Fall 2025

- Sole instructor for 21 students; designed and delivered the syllabus, lectures, assignments, examinations, and assessment.
- Taught logic, sets, combinatorics, and mathematical reasoning; incorporated group activities and an HTML personal-website project.

Extended Teaching Assistant – CISC 1100/1400
Fordham University
Fall 2024–Spring 2025

- Delivered approximately three to five class sessions under faculty supervision; explained homework solutions and supported grading.
- Led project-support sessions for students developing personal websites in HTML.

Teaching Assistant – Discrete Structures (CISC 2100)
Fordham University
Fall 2026–present

- Support problem-solving instruction, student questions, and assessment of coursework and examinations.

Research Mentoring

Summer Research Mentor – Federated Learning for WAVN
Fordham University
Summer 2025

- Mentored three undergraduate and master's students conducting research on federated learning for Wide Area Visual Navigation.

Publications

Peer-Reviewed Journal Articles

- **N. Paykari**, D.M. Lyons, and M. Rahouti. "Decentralized, Distributed, and Resilient Pathfinding for Dynamic, GPS-Denied Environments Using Visual Common Landmarks." *ACM Transactions on Cyber-Physical Systems (TCPS)*, vol. 10, no. 4, pp. 1–33, 2026.
- **N. Paykari**, M. Rahouti, and D.M. Lyons. "A Novel Blockchain-Driven Proof-of-Stake Model for Cooperative Navigation in Visual Homing Robotic Teams." *ACM Distributed Ledger Technologies: Research and Practice (DLT)*, vol. 5, no. 4, pp. 1–25, 2026.
- **N. Paykari**, D.M. Lyons, and M. Rahouti. "Enhancing Visual Homing in Robotics: A Study on Blockchain Integration and Consensus Algorithms." *ACM Distributed Ledger Technologies: Research and Practice (DLT)*, vol. 4, no. 1, pp. 1–27, 2025.

Peer-Reviewed Conference Proceedings

- **N. Paykari**, I. Vural, N. Desta, A. Zain, T. Clark, M. Rahouti, and D.M. Lyons. "Cross-View Landmark Recognition for Cooperative Visual Navigation." In *Proceedings of SPIE Defense + Commercial Sensing (Unmanned Systems Technology XXVIII)*, vol. 14047, article 1404707, 2026.

- N. Desta, I. Vural, **N. Paykari**, D.M. Lyons, and M. Rahouti. "Federated Visual Homing for Cooperative Multirobot Navigation in GPS-Denied Environments." In *Proceedings of SPIE Defense + Commercial Sensing (Unmanned Systems Technology XXVIII)*, vol. 14047, article 1404709, 2026.
- **N. Paykari**, T. Clark, A. Zain, D.M. Lyons, and M. Rahouti. "Witness Byzantine Fault Tolerance with Signature Tree and Proof-of-Navigation for Wide Area Visual Navigation." In *Proceedings of the 22nd International Conference on Informatics in Control, Automation and Robotics (ICINCO)*, vol. 2, pp. 633–644, 2025.
- **N. Paykari**, A. Alfatemi, D.M. Lyons, and M. Rahouti. "Integrating Robotic Navigation with Blockchain: A Novel PoS-Based Approach for Heterogeneous Robotic Teams." In *Proceedings of IEEE International Conference on Ubiquitous Robots (UR)*, 2024. *Best Paper Award Finalist*
- **N. Paykari**, D.M. Lyons, and M. Rahouti. "RoboStake: Pioneering Cooperative Navigation with a Novel Blockchain-Powered Proof-of-Stake in Robotic Teams." In *Proceedings of IEEE International Conference on Omni-layer Intelligent Systems (COINS)*, 2024.
- A. Alfatemi, S.A.L. Jamal, **N. Paykari**, M. Rahouti, and A. Chehri. "Multi-Label Classification with Deep Learning and Manual Data Collection for Identifying Similar Bird Species." *Proceedings of KES 2024, Procedia Computer Science*, vol. 246, pp. 558–565, 2024.
- A. Alfatemi, S.A.L. Jamal, **N. Paykari**, M. Rahouti, R. Amin, and A. Chehri. "Refining Bird Species Identification through GAN-Enhanced Data Augmentation and Deep Learning Models." *Proceedings of KES 2024, Procedia Computer Science*, vol. 246, pp. 548–557, 2024.
- **N. Paykari**, D.M. Lyons, and M. Rahouti. "Assessing Blockchain Consensus in Robotics: A Visual Homing Approach." In *Proceedings of IEEE 14th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)*, pp. 577–583, 2023.
- **N. Paykari**, S.H. Abbasi, and F. Shabaninia. "Design of MIMO Mamdani Fuzzy Logic Controllers for Wall Following Mobile Robot." In *Soft Computing Applications: Proceedings of the 5th International Workshop (SOFA)*, Advances in Intelligent Systems and Computing, Springer, 2013.

Manuscripts Under Review

- **N. Paykari**, D.M. Lyons, M. Rahouti, and A. Chehri. "From Shared Sight to Verified Waypoints: Blockchain-Ordered Navigation for Internet of Robotic Things." Under review, *IEEE Internet of Things Magazine*.
- "Robust Image-Guided Path-Planning Using a Decentralized Cooperative Consensus." Under review, *IEEE Transactions on Industrial Informatics*.

Preprints

- **N. Paykari**, R. Jokar, A. Alfatemi, D. Lyons, and M. Rahouti. "Optimizing Control Strategies for Wheeled Mobile Robots Using Fuzzy Type I and II Controllers and Parallel Distributed Compensation." *arXiv:2409.17161*, 2024. https://doi.org/10.48550/arXiv.2409.17161.

Presentations

- "Cross-View Landmark Recognition for Cooperative Visual Navigation." SPIE Defense + Commercial Sensing, 2026.
- "Federated Visual Homing for Cooperative Multirobot Navigation in GPS-Denied Environments." SPIE Defense + Commercial Sensing, 2026.
- "Integrating Blockchain and Wide Area Visual Navigation." Three Minute Thesis (3MT), Fordham University, 2025.
- Poster on wide-area visual navigation. NYU Vision Day, New York University, 2025.

- “Integrating Robotic Navigation with Blockchain: A Novel PoS-Based Approach for Heterogeneous Robotic Teams.” IEEE International Conference on Ubiquitous Robots (UR), poster presentation, 2024.
- “RoboStake: Pioneering Cooperative Navigation with a Novel Blockchain-Powered Proof-of-Stake in Robotic Teams.” IEEE International Conference on Omni-layer Intelligent Systems (COINS), online presentation, 2024.
- Poster on wide-area visual navigation. Research Day, Fordham University, 2024.
- “Assessing Blockchain Consensus in Robotics: A Visual Homing Approach.” IEEE UEMCON, 2023.

Honors and Professional Development

- Best Paper Award Finalist, IEEE International Conference on Ubiquitous Robots (UR), 2024.
- Preparing Future Faculty Intensive Program (PFFI), Fordham University, 2024–2025. Certificate of completion, May 2, 2025.
- Focus on Peer Review, Nature Masterclasses, November 8, 2024. Completed 3.5 hours of training.

Additional Research Appointments

Summer Professional Development Position – Robotics and Blockchain
Fordham University
Summer 2024

Summer Professional Development Position – Fordham–IBM Research
Fordham University
Summer 2023

Industrial Experience

Senior Control Systems Supervisor
Petropars Operation and Management Company (POMC)
2019–2021

- Supervised more than 15 engineering and technical staff in control-system implementation, testing, commissioning, and operational handover for petroleum facilities.
- Led site acceptance testing, loop checks, functional testing, and commissioning of distributed control, emergency shutdown, fire and gas, and power distribution control systems.
- Integrated industrial communication across Siemens and Yokogawa systems using TCP/IP, Modbus RTU/TCP, and fieldbus networks; supported client/server information exchange and shared hardware resources.
- Completed unfinished control-package logic, diagnosed hardware/software integration problems, and supported functional verification and startup.
- Worked with Siemens PCS7 and Yokogawa Centum platforms and HIMA safety systems to validate control logic and cause-and-effect sequences.

Control Systems Supervisor
Petropars Operation and Management Company (POMC)
2015–2019

- Supervised site acceptance testing, loop testing, and pre-commissioning, commissioning, and startup of DCS, ESD, and fire and gas systems.
- Coordinated technicians, prepared commissioning procedures, resolved field technical queries, and verified system readiness.
- Diagnosed control and instrumentation faults across field devices, cabinets, communication networks, and operator interfaces.

Control and Instrumentation Engineer
PARSIAN Group, Electrical and Control Department
2012–2015

- Engineered and commissioned power distribution control systems (PDCS) covering approximately 25,000 physical and soft I/O points.
- Coordinated five to eight vendor and client personnel during engineering, integration, testing, and commissioning activities.
- Developed system architectures, I/O lists, logic and wiring diagrams, control narratives, cause-and-effect matrices, and cabinet documentation.
- Programmed PLC/DCS control and interlocking logic; designed HMI/SCADA operator interfaces using Siemens WinCC.
- Integrated industrial networks and supported hardware/software testing, acceptance, and handover.

Safety Systems and Instrumentation Engineer
Azar Imen Persian
2010–2012

- Designed and documented control and safety-system components, including I/O lists, logic diagrams, wiring diagrams, and control philosophies.
- Supported hardware design, panel assembly, configuration, testing, and commissioning of conventional (zone-based) and addressable fire alarm systems, including panels and equipment.
- Applied fire-alarm design and testing practices informed by NFPA 72 (National Fire Alarm and Signaling Code).
- Performed fault diagnosis and acceptance testing; prepared as-built documentation and handover materials.

Technical Skills

- **Programming and analysis:** Python, C++, MATLAB/Simulink; PLC programming using ladder logic (LAD), function block diagrams (FBD), Structured Control Language (SCL), and Statement List (STL).
- **Robotics:** ROS, Gazebo, visual homing, cooperative navigation, cross-view landmark recognition, topological graph pathfinding.
- **Distributed systems:** Blockchain consensus, Proof of Stake, Byzantine fault tolerance, industrial client/server systems.
- **Automation platforms:** Siemens PCS7, STEP 7, TIA Portal, WinCC; Yokogawa Centum CS3000/VP; HIMA safety PLCs; DCS, PDCS, PLC, SCADA, HMI, ESD, and fire and gas systems.
- **Fire and gas alarm systems:** Fire and Gas Systems (FGS), conventional (zone-based) and addressable fire alarm panels, mimic panel, alarm logic, testing, and commissioning.

- **Industrial networking:** Modbus RTU/TCP, TCP/IP, PROFIBUS DP/PA, PROFINET, Industrial Ethernet, OPC UA/DA, and remote terminal units.
- **Cyber-physical systems:** Integration of computation, communication, and physical processes in robotic and industrial control systems; Byzantine fault-tolerant coordination for cooperative robotic teams.
- **Engineering and validation:** Hardware/software integration, control logic, instrumentation, I/O mapping, loop checks, factory and site acceptance testing, commissioning, and technical documentation.

Professional Service

Journal reviewing: ACM Transactions on Cyber-Physical Systems; IEEE Internet of Things Journal; IEEE Transactions on Industrial Informatics; Information Fusion; IEEE Transactions on Dependable and Secure Computing.

Conference service and reviewing: IEEE International Conference on Tools with Artificial Intelligence (ICTAI); International Conference on Informatics in Control, Automation and Robotics (ICINCO).

Languages

Kurdish: native; Persian: native; English: full professional working proficiency.

Academic References

Damian M. Lyons – Ph.D. co-adviser
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